

Kanchana Herath

ssociate Software Quality Assurance Engineer

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PROFESSIONAL SUMMARY

Detail-oriented Software Quality Assurance Engineer with experience in manual testing, database validation, and defect management in Agile environments. Skilled in delivering high-quality software through structured testing and analytical problem-solving. Developed and deployed a production-ready multi-role bus booking platform (goJourney), demonstrating full-stack development expertise and a strong focus on quality, performance, and user experience.

PROFESSIONAL EXPERIENCE

01/2026 – 07/2026
Colombo, Sri Lanka

LAYOUTindex Pvt. Ltd, Colombo, Sri Lanka
Software Quality Assurance Engineer Intern

- Performed functional, regression, and end-to-end testing for enterprise applications across Web, POS, Kiosk, and Mobile platforms.
- Validated system functionality against business requirements and identified deviations through structured testing.
- Conducted SQL database validation to verify data accuracy, integrity, and consistency between application and database layers.
- Identified, documented, tracked, and verified defects while supporting timely issue resolution.
- Analyzed business workflows to ensure system processes aligned with functional requirements.
- Prepared detailed test cases, defect reports, test execution reports, and release validation documentation.
- Collaborated with developers, business stakeholders, and support teams throughout Agile development cycles.
- Assisted in investigating production issues through root cause analysis and system validation.

PROJECT EXPERIENCE

goJourney
Full-Stack Bus Booking Platform (Featured Project)

- Architected, developed, and deployed a production-grade Progressive Web Application (PWA) supporting passengers, bus operators, and administrators.
- Implemented real-time seat booking with conflict resolution to prevent double-booking under concurrent usage.
- Integrated PayHere payment gateway with automated booking confirmation and subscription billing system.
- Developed live GPS tracking functionality using Firebase Realtime Database and Leaflet.js.
- Built multi-language support (Sinhala, Tamil, English) with responsive UI across all devices.
- Applied strong security practices including role-based access control, Firestore security rules, and HTTPS enforcement.
- Conducted extensive QA including cross-browser testing, responsive testing (320px–2560px), and edge-case validation.

QA Testing Projects
E2E Testings and Migration Testings

- Conducted End-to-End, Regression, and Migration Testing for multiple enterprise applications including Alittle, INDEEPA, Carousel, IDHIKR App, and WASHAPP.
- Verified data migration accuracy and system functionality following application upgrades.
- Reported findings, validated fixes, and maintained testing documentation to support production releases.

Academic Project – Lecturer and Students Evaluation System
- Performed manual and end-to-end testing for CRUD operations.
- Validated data consistency between Visual Basic frontend and MySQL database.
- Ensured accuracy of evaluation calculations and data integrity.

EDUCATION

06/2022 – 11/2025
Batticaloa

BSc. Physical Science (Computer Science)
Eastern University Sri Lanka
GPA: 2.96

2020

G.C.E. Advanced Level — Physical Science
PL/Giritale Gama National College
Physics: C, Mathematics: S, Chemistry: S

2016

G.C.E. Ordinary Level
PL/Giritale Gama National College
7A Passes, 1B, 1C, 9 Subjects

TECHNICAL SKILLS

IT Audit & Data Analysis
- SQL
- Data Validation
- Business Process Validation
- Root Cause Analysis
- Risk Identification
- Process Documentation

Testing & Quality Assurance
- Functional Testing
- Regression Testing
- End-to-End Testing
- User Acceptance Testing (UAT)
- Cross-Browser Testing
- Cypress Automation (Basic)

Databases
- Firebase (Firestore, RTDB)
- MySQL, MongoDB

Tools & Platforms
- Git / GitHub
- Vercel, VS Code, PyCharm
- Browser DevTools
- MS Office

Core Concepts
- SQA / STLC / Agile
- CRUD & Data Validation
- Data Structures & Algorithms
- REST APIs & PWA

PROFESSIONAL SKILLS

Analytical Thinking
Strong ability to identify edge cases and logical flaws; applied in implementing String Matching Algorithms and goJourney's concurrent seat-lock system.

SDLC / STLC Awareness
Practical understanding of software development and testing life cycles, applied daily during internship in an Agile delivery environment.

Attention to Detail
Committed to high standards evidenced by zero post-launch defects on ZONO TIMER and production-grade security implementation in goJourney.

Documentation & Communication
Skilled at writing clear test cases, defect reports, and technical documentation to bridge QA and development teams effectively.

LANGUAGES

Sinhala

English

REFERENCES

M.K.D.K.R Perera , *Tech Lead,* LAYOUTIndex Pvt. Ltd
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